



STRATHMORE DATA COMMUNITY

DataSprint 2026 — Problem Statement

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Background

Kenya has made significant strides in financial inclusion over the past decade. Mobile money has transformed how millions of Kenyans save, borrow, and transact, yet the picture on the ground remains uneven. According to the 2024 FinAccess Household Survey, conducted by the Central Bank of Kenya, KNBS, and FSD Kenya, 9.9% of Kenyan adults remain fully excluded from financial services. More significantly, over half of those surveyed (52.6%) reported that their financial situation has worsened compared to the previous year.

Behind every data point is a real person; a smallholder farmer who could not repay a loan after a drought, a young woman in Turkana with no national ID and therefore no bank account, a household in Nairobi spending more than it earns. Understanding who is struggling financially and why — is one of the most important questions facing Kenya’s policymakers, banks, NGOs, and development partners today.

The Challenge

Using data from 20,871 Kenyan adults surveyed in 2024, build a machine learning model that predicts whether a person’s financial situation has Improved, Stayed the same, or Worsened, and identify the key factors that drive financial outcomes in Kenya.

This is a multiclass classification problem. Your model will predict one of three outcomes for each individual:

- Improved — the person’s financial situation is better than the previous year
- Stayed the same — no significant change in financial situation
- Worsened — the person’s financial situation has deteriorated

The Dataset

The dataset has been curated by iLabAfrica (Kevin Obote) from the 2024 FinAccess Household Survey — the most recent and comprehensive survey of financial inclusion in Kenya, published in December 2024. It covers 20,871 individuals across all 47 counties.

Demographics

Column	Description	Example Values
county	Kenya county of respondent	Nairobi City, Mombasa, Kisumu, Turkana
location_type	Urban or rural setting	Urban, Rural
Sex	Gender of respondent	Male, Female
Age	Age group	18–25, 26–35, 36–45, 46–55, Above 55
household_size	Number of people in household	1 to 15+
education_level	Highest education completed	None, Primary, Secondary, Tertiary
marital_status	Marital status	Married, Single, Widowed, Divorced
has_disability	Disability status	With Disability, Without Disability

Livelihood & Income

Column	Description	Example Values
monthly_income	Total income received in past month (KES)	1000, 5000, 30000
experienced_shock	Experienced a financial shock in past year	Yes, No

Mobile & Digital Access

Column	Description	Example Values
mobile_ownership_1	Owns a mobile phone	Yes, No
mobile_money_access	Has ever accessed mobile money	Yes, No
barriers_mobile_money	Main barrier to mobile money access	0 (none), Phone ownership, Eligibility, Affordability

Financial Behaviour

Column	Description	Example Values
Savings_formal	Uses formal savings (bank, SACCO, MFI)	Usage, Non-usage
Savings_informal	Uses informal savings (chama, group)	Usage, Non-usage
Loan_formal	Has a formal loan	Usage, Non-usage
Loan_informal	Has an informal loan	Usage, Non-usage
defaulted	Has defaulted on a loan or debt	Yes, No
formal_service_use	Uses any formal financial service	Usage, Non-usage
barriers_bank	Main barrier to bank access	Affordability, Eligibility, Awareness, Access, Trust
prodsum1	Number of financial services used (count)	0 to 10+

Financial Health & Literacy

Column	Description	Example Values
fl_score	Financial literacy score	None correct, One correct, Two correct, All correct
nfhi_11	Was food secure in past 12 months	Yes, No
nfhi_12	Managed non-food spending adequately	Yes, No
nfhi_13	No debt stress in past 3 months	Yes, No
accessto_13k_1month	Can access KES 13,000 emergency funds in 1 month	Yes, No
not_difficult	Emergency funds not difficult to access in 30 days	Yes, No

Target Variable

Column	Description	Class Distribution
financial_status	Financial situation vs. previous year	Worsened: 52.6% Stayed the same: 26.9% Improved: 20.5%

The Guiding Question

Beyond building a model, your team must answer this question:

Which factors most strongly predict financial deterioration among Kenyan adults, and what does your model suggest policymakers, banks, or NGOs should prioritise to improve financial wellbeing?

What You Are Expected To Do

Step	Task
1. Data Cleaning	Handle missing values, check for inconsistencies, understand each feature
2. EDA	Visualise distributions, explore relationships between features and the target, generate insights
3. Preprocessing	Encode categorical variables, scale numerical features, prepare data for modelling
4. Modelling	Train at least one classification model that predicts financial_status
5. Evaluation	Use weighted F1-score as your primary metric, report performance on a held-out test set
6. Interpretation	Identify important features and translate your findings into actionable insights

Deliverables

Deliverable	Requirements
Jupyter Notebook (.ipynb)	Full solution — data cleaning, EDA, preprocessing, model training, evaluation, and interpretation. Must be clean, well-commented, and reproducible.
Data Visualisations	Minimum 5 meaningful charts included in the notebook. Well-labelled and interpretable by a non-technical audience.
7-Slide PowerPoint (.pptx)	Slides: (1) Problem, (2) Dataset & EDA, (3) Key EDA finding, (4) Modelling approach, (5) Model performance, (6) Key drivers — answer to guiding question, (7) Recommendations.

Evaluation Metric & Dataset Access

Your model is evaluated using the Weighted F1-Score, which accounts for class imbalance. A model that only predicts ‘Worsened’ will score poorly because it fails on ‘Improved’ and ‘Stayed the same’.

Dataset (Kaggle — recommended): <https://www.kaggle.com/datasets/davidpbriggs/kenya-finaccess-household-survey-2024>

Official FinAccess Report & Manual: <https://finaccess.knbs.or.ke/reports-and-datasets>

Note on Missing Values

Column	Missing %	How to Handle
barriers_bank	27.5%	These are respondents who have a bank — no barrier applies. Impute with a new category: ‘No barrier’ or 0.
monthly_income	~0% (pre-imputed)	Already imputed with median in the curated dataset.

Contact

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Learn. Build. Launch.

The Strathmore Data Community is an officially recognised student community at Strathmore University, Nairobi.